Sentiment Classification of Movie Reviews Using Bidirectional LSTM & NLP

TensorFlow / Keras | Bidirectional LSTM | IMDb Dataset | NLP Pipeline | Binary Classification

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01 PROBLEM STATEMENT

- Digital content such as movie reviews, social media posts, and product feedback is growing rapidly.
- Automated systems are needed to understand human emotion and opinion in text.
- Manually classifying thousands of reviews is impractical and time-consuming.
- This project focuses on **binary sentiment classification** of movie reviews.
- The goal is to predict whether a review expresses **Positive** or **Negative** sentiment using a deep learning NLP pipeline.

> *"Can a Bidirectional LSTM model, trained end-to-end on raw text, learn the nuanced linguistic patterns that distinguish praise from criticism in movie reviews?"*

02 DATASET

IMDB LARGE MOVIE REVIEW DATASET

50K Labeled Reviews

50/50 Pos / Neg Balance

233 Avg. Words/Review

2,470 Max Review Length

● Train 20K ● Test 25K ● Val 5K

30.2% of reviews exceed the 250-token sequence limit; shorter reviews are zero-padded to maintain uniform input dimensions.

03 METHODOLOGY & NLP PIPELINE

Raw Text Input — Movie reviews as plain text strings
→ **TextVec-torization** — Vocab: 20K tokens · Seq len: 250
→ **Embedding Layer** — 20001 vocab · dim=32 · SpatialDrop.1
→ **Bi-LSTM Stack ×2** — 64 units→32 units· bidirectional context
→ **Dense + Dropout** — 64 ReLU units · rate=0.5
→ **Sigmoid Output** — P(positive) ∈ [0,1]

04 MODEL ARCHITECTURE

`Embedding`	`(None, 250, 32)`	Maps each token to a 32-dim dense vector
`SpatialDropout1D`	`(None, 250, 32)`	Drops entire embedding feature maps (rate=0.3)
`Bi-LSTM (64)`	`(None, 250, 128)`	return_sequences=True – captures full context
`Bi-LSTM (32)`	`(None, 64)`	Condenses sequence into one context vector
`Dense (64, ReLU)`	`(None, 64)`	Non-linear feature transformation
`Dropout (0.5)`	`(None, 64)`	Prevents co-adaptation during training
`Dense (Sigmoid)`	`(None, 1)`	Binary probability output

05 TRAINING CONFIG

OPTIMIZER **Adam**	LR **0.001**
LOSS **BinaryCE**	BATCH **32**
MAX EPOCHS **20**	PATIENCE **5**
VOCAB SIZE **20,000**	EMBED DIM **32**

Early Stopping monitors val_loss and restores best weights automatically to prevent overfitting.

APPROACH COMPARISON

Approach	IMDb Acc.
Lexicon-based	~72%
ML (SVM/NB)	~83%
Unidir. LSTM	~89%
BiLSTM (ours)	**83.5%**
BERT / Transformer	~94%

06 RESULTS & DISCUSSION

83.5% TEST ACCURACY

0.3965 TEST LOSS

Class	Prec.	Recall	F1
NEG	0.84	0.83	0.83
POS	0.83	0.84	0.84
Macro Avg	**0.84**	**0.84**	**0.83**

Balanced precision & recall across both classes confirms the model is unbiased — a direct result of the perfectly balanced 50/50 dataset.

KEY FINDINGS & DISCUSSION

- ✅ **Strong balanced performance:** F1-scores of 0.83–0.84 for both classes show the model is well-calibrated and does not favour either sentiment.
- ✅ **High-confidence predictions:** Clearly positive reviews (e.g., "One of the best movies I've ever seen") scored >90% confidence; clearly negative reviews scored similarly high.
- ⚠️ **Double-negation misclassification:** The phrase "not bad" was labelled NEGATIVE (73.2% confidence) because the model associated "not" with negative contexts during training.
- ⚠️ **Mild overfitting:** Training accuracy of 88–90% vs. test accuracy of 83.5% — managed effectively by EarlyStopping.
- → **Future Work:** Replace learned embeddings with GloVe/FastText; fine-tune BERT for expected ~93% accuracy; implement dedicated negation preprocessing.